

Feature Engineering, Model Optimization & Performance Comparison

AIML Internship – Task 2

1. Introduction

The objective of this project was to improve the performance of a house price prediction system using the California Housing Dataset. Unlike the previous task, this project focused on feature engineering, model optimization, and performance comparison among multiple regression algorithms. The goal was to identify the most suitable model for predicting house prices accurately.

The California Housing Dataset contains information related to housing districts, including median income, house age, average rooms, population, occupancy, latitude, and longitude. These features were used to predict the median house value.

2. Methodology

Dataset Preparation

The California Housing Dataset was loaded using the Scikit-Learn library. Initial data exploration was performed to understand the structure of the dataset, verify data types, and check for missing values. Statistical summaries and correlation analysis were also conducted to gain insights into feature relationships.

Feature Engineering

Feature scaling was applied using the StandardScaler technique. Since different features have different value ranges, scaling ensured that all features contributed equally during model training. This preprocessing step improves model stability and performance.

Model Development

The dataset was divided into training and testing sets using an 80:20 ratio. Three regression algorithms were implemented and evaluated:

1. Linear Regression
2. Ridge Regression
3. Decision Tree Regressor

These models were selected to compare the performance of traditional linear methods with a non-linear tree-based approach.

Evaluation Metrics

Model performance was evaluated using:

- Root Mean Squared Error (RMSE)
- R^2 Score (Coefficient of Determination)

RMSE measures prediction error, while R^2 Score indicates how well the model explains variations in house prices.

3. Results and Performance Comparison

Three regression models were trained and evaluated using the California Housing Dataset. The performance of each model was measured using Root Mean Squared Error (RMSE) and R^2 Score.

Model Comparison Table

Model	RMSE	R^2 Score
Decision Tree Regressor	0.724234	0.599732
Ridge Regression	0.745554	0.575819
Linear Regression	0.745581	0.575788

Analysis of Results

The comparison results indicate that the **Decision Tree Regressor** achieved the best overall performance among the three models.

- The Decision Tree model obtained the **lowest RMSE (0.724234)**, indicating lower prediction error compared to the other models.
- It also achieved the **highest R^2 Score (0.599732)**, meaning it explained approximately 60% of the variance in house prices.
- Ridge Regression and Linear Regression produced very similar results, with nearly identical RMSE and R^2 values.
- The slightly better performance of the Decision Tree model suggests that it was able to capture non-linear relationships in the housing data more effectively than the linear models.

Visual Analysis

Additional visualizations such as:

- RMSE Comparison Chart
- R^2 Score Comparison Chart
- Actual vs Predicted Plot
- Residual Distribution Plot
- Feature Importance Plot

were generated to further evaluate model behavior and prediction accuracy.

Based on both numerical evaluation and graphical analysis, the **Decision Tree Regressor** was selected as the best-performing model and saved for future predictions.

4. Conclusion

This project successfully demonstrated feature engineering, model optimization, and performance comparison using the California Housing Dataset.

Three regression models—Linear Regression, Ridge Regression, and Decision Tree Regressor—were trained and evaluated after applying feature scaling. Performance was assessed using RMSE and R^2 Score.

Among all models, the **Decision Tree Regressor** achieved the best results with:

- **RMSE = 0.724234**
- **R^2 Score = 0.599732**

These results indicate that the Decision Tree model provided more accurate predictions and better captured complex patterns in the housing data compared to the linear models.

The project enhanced understanding of data preprocessing, feature scaling, model evaluation, and model selection techniques that are widely used in real-world machine learning applications.

Future Improvements

- Hyperparameter tuning using GridSearchCV
- Random Forest Regression
- XGBoost Regression
- Cross-validation techniques
- Deployment of the trained model as a web application

This project provided practical experience in building, evaluating, and comparing machine learning models for regression-based prediction problems.